



Algorithmic trading module Light

User's manual

Version 4.3

The Quik logo, which consists of the word "quik" in a white, lowercase, sans-serif font, centered within a solid green circle.

quik

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1. General

Algorithmic orders (or algo orders) is a special type of orders with conditions preset by a previously developed algorithm. Such algorithms are processed in a special module of the QUIK server. Once an event covered by the algorithm occurs, the module sends into the trading system transactions that require entry or cancellation of orders hereinafter referred to as **linked orders**.

The algorithmic orders' module does not check the adequacy of the client's funds for an algo order being entered. The limits are checked for each linked order. If the client's funds are sufficient, an algo order proceeds with generating linked orders until its work is done. Otherwise, it stops as soon as the available funds are exhausted. If a linked order cannot be entered into or is rejected by the trading system, the algo order will be cancelled.

Monitoring the algorithmic orders execution status is available from the QUIK workstation using special tables:

- [Algo orders table](#) contains the list of entered algorithmic orders and allows handling them;
- [Linked orders table](#) contains the list of orders that have been entered into the trading system as a result of executing the selected algorithmic order;
- [Linked trades table](#) contains the list of trades that have been carried out as a result of filling linked orders.

The Module supports the following types of algorithmic orders:

- [Volatility](#);
- [GTD/GTC](#) orders;
- [Stop order](#);
- [Spread](#).

Each algorithm is provided with an individual dialog box for entry of the required algo order parameters. Each broker may have different lists of available algorithmic orders. For availability of order types described below please consult your broker.

Functions that allow handling algorithmic orders are available from the **Action** menu or the shortcut menu of the Algo orders table, if provided by the broker's server. In order to work with these functions, one should activate in the menu item **System / Data request / Available securities...** the reception of data by class relative to algorithmic order types.

Detailed information about the processing of an algorithmic order is available through the [request for report](#) shown in an individual window. It is also possible to [request a report on the algorithmic order for a previous period](#).

2. Table of algorithmic orders

menu **Create window/ Algo orders table**

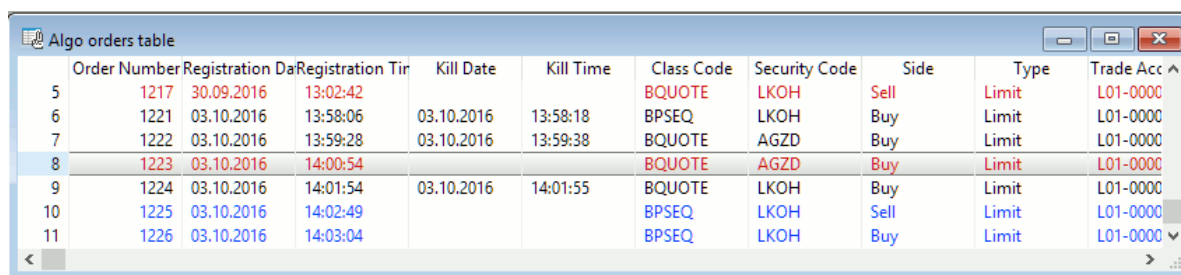
To display the **Algo orders table** command under the **Create window** menu, select **Create window/Edit menu...** and add the Algo orders table to the list of selected items (for details, see the User's Manual of the QUIK workstation, Section 2 "Basic Operating Principles", p. 2.15.2).

2.1 Purpose of the table

This table is intended for displaying algorithmic orders and their status. The table allows execution of standard functions such as entry/cancellation of algo orders and generation of reports.

2.2 Table format

The table contains a chronologically arranged list of algorithmic orders. Each algo order is represented by a line in the table. Each parameter is shown in an individual column.



	Order Number	Registration Date	Registration Time	Kill Date	Kill Time	Class Code	Security Code	Side	Type	Trade Acc
5	1217	30.09.2016	13:02:42			BQUOTE	LKOH	Sell	Limit	L01-0000
6	1221	03.10.2016	13:58:06	03.10.2016	13:58:18	BPSEQ	LKOH	Buy	Limit	L01-0000
7	1222	03.10.2016	13:59:28	03.10.2016	13:59:38	BQUOTE	AGZD	Buy	Limit	L01-0000
8	1223	03.10.2016	14:00:54			BQUOTE	AGZD	Buy	Limit	L01-0000
9	1224	03.10.2016	14:01:54	03.10.2016	14:01:55	BQUOTE	LKOH	Buy	Limit	L01-0000
10	1225	03.10.2016	14:02:49			BPSEQ	LKOH	Sell	Limit	L01-0000
11	1226	03.10.2016	14:03:04			BPSEQ	LKOH	Buy	Limit	L01-0000

Parameters of algo orders shown in table columns are described below:

Parameter	Description
Order Number	Algo order registration number
** Registration Date	Algo order registration date
** Registration Time	Time of registering an algo order at the QUIK server
** Registration Time (mcs)	Time of registering an algo order at the QUIK server, in mcs
** Kill date	Date of algo order cancelling on QUIK server
** Kill Time	Time of cancelling an algo order at the QUIK server
** Kill Time (mcs)	Time of cancelling an algo order at the QUIK server, in mcs
*** Security code	Financial instrument ID
*** Security name	Short name of an instrument in the trading system

Parameter	Description
*** Security Full Name	Full name of an instrument in the trading system
Side	Trade direction: Buy/Sell
*** Trade account	Trading account number
Firm ID	ID of a trading participant on whose behalf the algo order has been entered
Quantity	Total quantity of securities expressed in lots. For algo orders of Stop order type: quantity in active linked order
Qty in Iteration	Lot quantity to be fulfilled during the one iteration. The parameter is not filled if the value of the Qty/Value parameter is Value. This parameter is applicable to Spread algo orders
Value	Volume of an algo order in terms of cash (excluding the exchange commission and accrued income). This parameter is applicable to GTD/GTC algo orders
Value in Iteration	Volume the linked algo orders are registered for. The parameter is not filled if the value of the Qty/Value parameter is Qty. This parameter is applicable to Spread algo orders
Currency	Currency in which the volume is set. This parameter is applicable to Spread algo orders
*** Value Balance	Volume balance. This parameter is applicable to Spread algo orders
*** Executed Quantity	Executed part of an algo order expressed in lots
*** Executed Value	Executed part of an algo order in terms of cash
Execution percent	– Execution percent = Executed Quantity / Quantity * 100%, if order size is measured in lots; – Execution percent = Executed Value / Value * 100%, if order size is measured in cash. This parameter is applicable to GTD/GTC algo orders
Time in Force	Period of time in which an algo order should be filled. This parameter is applicable to GTD/GTC, Stop, Spread algo orders Valid values for GTD/GTC algo orders: – Good till date; – Good till time; – Good till cancel – the execution period is unlimited. Stop orders may have any of the following values: – Session; – Good till time; – Good till time; – Good till cancel Spread algo orders may have any of the following values: – Session; – Good till time; – Good till time; – Good till cancel

Parameter	Description
** Expiry Date	Date till which (inclusive) an algo order should be filled. This parameter is applicable to GTD/GTC algo orders, Spread and Stop orders
*** Client Code	Client code at the QUIK server
* Status	Algo order status (Active, Filled, Cancelling, Cancelled)
Settle code	Code of settlements
Current session	Indicates that order was sent in current session: Yes / No
Algorithm	Type of algorithmic order, e.g.: Stop order
*** Class Code	Instrument class code in the trading system
*** Class Name	Instrument class name
*** Firm ID of Linked Orders	ID of a firm participating in trade on whose behalf the linked algo order has been entered
Balance	Unfilled part of an algo order expressed in lots
Price	Price per unit of an instrument shown in algo order
*** Execution VWAP	Weighted average execution price calculated for linked trades in cash
*** Execution VWAP (price)	Weighted average execution price calculated for linked trades in pips (for derivatives market securities)
User	User's ID (UID) in QUIK
Memo	Textual comment to an algo order. Linked orders and trades will contain this comment
Cancel reason	Reason for algo order cancelling
Last Trade Price	The price of the last linked trade
Execution alerts	Indicates that alerts about linked orders execution are sent
Internal Comment	Text comment for internal use (is not sent to the trading system)
Error Text	The reason for a linked order rejection by the QUIK server
Client Qualifier (MiFIDII)	Client Qualifier according to MiFID II directive. Valid values: _ 0 – No qualifier; _ 1 – Natural person; _ 3 – Legal entity
Client ID (MiFIDII)	Client identifier according to MiFID II directive
Type	Algo order type: Market or Limit. This parameter is applicable to GTD/GTC algo orders, Stop orders

Parameter	Description
Volatility	Volatility value. This parameter is applicable to Volatility orders
Volatility Deviation	Allowable deviation of volatility. This parameter is applicable to Volatility orders
Type of Volatility Deviation	_ Absolute – volatility deviation is set in absolute items; _ % from volatility – volatility deviation is set in percent This parameter is applicable to Volatility orders
** Start time	Time of starting an algorithmic order during the day
** Stop time	Time of terminating an algorithmic order during the day
** Expiry time	Expiration time of an algo order. An algo order is cancelled upon occurrence of this time in expiration date. The parameter is applicable for GTD/GTC algo orders, Stop orders
Iterations Number	Number of iterations to be done by an algorithmic order
Iteration Time	Duration of an iteration
Current Iteration Number	Number of the iteration currently done
Time Type	Algorithm start time: _ Set by user – from the time, set by user; _ Current – from the current time
Balance (Value)	Unfilled value of an algo order in monetary terms
Balance (Quantity)	Unfilled value of an algo order in lots
Qty/Value	Indicates how to display the order volume. Qty – in lots (Qty), Value – in monetary terms (V)
Use Work Interval	Attribute of the time interval use. Valid values: _ Yes – time interval is used; _ No – time interval is not used. The parameter is applicable for GTD/GTC algo orders, Stop orders, Spread
Safe Spread percent	Value of the protective spread in percent. The parameter is not filled if the protective spread type is 'Money' or 'Not set'. This parameter is applicable to Stop algo orders
Stop-limit percent	Value of Stop-limit in percent. The parameter is not filled if the protective spread type is 'Money' or 'Not set'. This parameter is applicable to Stop algo orders
Take-profit percent	Value of take-profit in percent. The parameter is not filled if the protective spread type is 'Money' or 'Not set'. This parameter is applicable to Stop algo orders
Limit kind	Type of limit. This parameter is applicable to Stop algo orders

Parameter	Description
Cancel Client Orders	Attribute of cancelling client orders when closing a position (Yes / No). This parameter is applicable to Stop algo orders
Stop-limit Type	_ Money – Stop-limit percent parameter is not filled; _ Not set – Stop-limit and Stop-limit percent parameters are not filled; _ Percent – Stop-limit parameter is not filled. This parameter is applicable to Stop algo orders
Take-profit Type	_ Money – Take-profit percent parameter is not filled; _ Not set – Take-profit and Take-profit percent parameters are not filled; _ Percent – Take-profit parameter is not filled. This parameter is applicable to Stop algo orders
Spread	Spread value. The value is shown in terms of cash or in the ratio depending on the value of the Safe Spread Type parameter. This parameter is applicable to Spread and Stop algo orders
Spread percent	Value of the protective spread in percent. The parameter is not filled if the protective spread type is 'Money' or 'Not set'. This parameter is applicable to Stop algo orders
Safe Spread Type	Valid values for Stop orders: _ Money – Spread percent parameter is not filled; _ Percent – Spread parameter is not filled. Valid values for Spread orders: _ Absolute – the value is shown in terms of cash in the Spread field; _ Relative – the value is shown in the ratio in the Spread field
Safe Spread	Value of the protective spread in price currency. The parameter is not filled if the protective spread type is Percent. This parameter is applicable to Stop algo orders
Stop-limit	Value of Stop-limit in price currency. The parameter is not filled if the Stop-limit type is 'Percent' or 'Not set'. This parameter is applicable to Stop algo orders
Take-profit	Value of Take-profit in price currency. The parameter is not filled if the Stop-limit type is 'Percent' or 'Not set'. This parameter is applicable to Stop algo orders

* – when cancelling an algo order it has **Cancelling** status till the Module receives all linked orders and trades. After all the required information is received the status of the algo order is changed to **Cancelled**.

** – if the **Show date and time of the trading data considering the local time zone** parameter is enabled (see QUIK User's Manual, Section 2, sub-section 2.10.1) then the value is displayed according to the time zone of the computer where the QUIK terminal is running.

*** – this field is not populated for Spread algo orders. The parameter is displayed independently for purchase and sale in two additional columns.

The order execution status is highlighted by different user-customizable font and background colors. The default font colors are:

- Blue, filled;
- Black, cancelled;
- Red, active;
- Grey, Cancelling.

2.3 Table settings

The table's edit dialog box contains the following settings:

1. **Table name** – for the table's heading.
2. **Rows** – for selection of types of algo orders to be shown in the table.
3. **Columns** – for selection of parameters to be shown in the table and determination of their sequence.
4. **State filter** – for display of algo orders relative to their execution status (Active, Filled, Cancelled, Cancelling).

5. **Operation filter** – for display of algo orders relative to their direction (Buy/Sell).
6. **Color settings** – for setting the line background color relative to order execution status. For more detail see [2.5](#).
7. **Clients filter** – selecting client codes to show them in the table.

2.4 Available functions

The table data are available for copying into the Windows Clipboard and exporting through the DDE server.

- Double left click of the mouse – to enter a new algo order with conditions similar to one on which the cursor is set;
- Double right click of the mouse – to cancel an active algo order on which the cursor is set;
- 'F2' – to enter a new order;
- 'Ctrl+D' – to withdraw an order;
- 'Ctrl+E' – to edit the table;
- 'Ctrl+W' – to automatically adjust the width of columns.

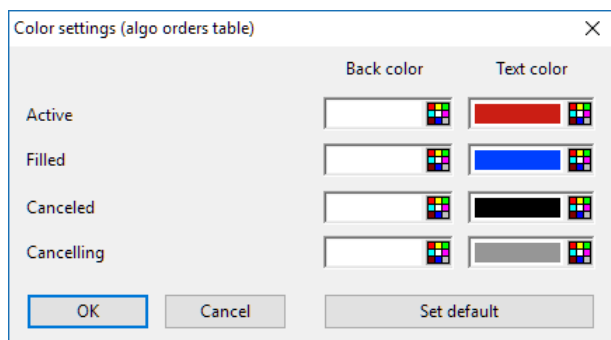
Functions available from the table's contextual menu or the **Action** program menu (if the table is active):

- **New algo order** – to open the dialog box used for entering a new algo order. If the type of a new algo order is the same as the type of the order selected in the table, than its parameters are filled automatically by the values of initial order. Otherwise the new algo order has only the same instrument as initial order have;
- **Cancel algo order** – to withdraw a highlighted algorithmic order from the trading system;
- **Cancel active orders** – withdraw all algorithmic orders from the trading system;
- **Resume algo order** – to resume the execution of highlighted algorithmic order with Cancelled status, if it is not expired (for GTD/GTC and Stop algo orders) or if it is sent during current trading session (for Volatility algo orders);
- [Get report](#) – to send a request for report regarding a selected algorithmic order, see [12.1](#);
- [Get report for previous period](#) – to send a request for report regarding an algorithmic order for a previous period, see [12.2](#);
- [Linked orders](#) – to open the **Linked orders. Algo order <Order number>** table to view orders that have been generated by the selected algo order, see p. [3](#). The “**Linked negotiated deal orders. Algo orders <Order number>**” table opens for linked orders put for NDM classes, see p. [4](#);
- [Linked trades](#) – to open the **Linked trades. Algo order <Order number>** table to view trades concluded from the selected algo order, see [5](#).

The complete list of control keys for all types of tables is shown in Annex to Section 2 of the QUIK User's Manual.

2.5 Color settings

The relevant window can be called by clicking on button '...' right of the **Color settings** checkbox in the edit dialog box of the Algo orders table. This function allows setting the background and font color for table lines relative to order execution status:



- **Active** – for algo orders whose status is Active;
- **Filled** – for algo orders whose status is Filled;
- **Cancelled** – for algo orders whose status is Cancelled;
- **Cancelling** – for algo orders whose status is Cancelling.

The button **Set default** returns the default settings shown in the picture.

3. Linked orders

3.1 Purpose of the table

The table **Linked orders. Algo order <Order number>** is intended for viewing orders generated by a certain algorithmic order that have been registered in the trading system in the current trading session.

3.2 Table format

The table contains a chronologically arranged list of linked orders. Each linked order is represented by a line in the table. Each parameter described in [3.3](#), pos. 8 is shown in an individual column.



	Number	Exchange cod	Sent(time)	Period	Security	Side	DEPO account	Price	Qt
23	1 192 730		18:46:08	Trading	LUKOIL [BK]	Sell	L01-00000F00	2 150,00	
24	1 192 732		18:46:14	Trading	LUKOIL [BK]	Sell	L01-00000F00	2 150,00	
25	1 192 734		18:46:22	Trading	LUKOIL [BK]	Sell	L01-00000F00	2 150,00	
26	1 192 735		18:46:33	Trading	LUKOIL [BK]	Sell	L01-00000F00	0,00	
27	1 192 736		18:46:33	Trading	LUKOIL [BK]	Sell	L01-00000F00	2 150,00	
28	1 192 737		18:47:03	Trading	LUKOIL [BK]	Sell	L01-00000F00	2 150,00	

3.3 Table settings

The table's edit dialog box contains the following settings:

1. **Table name** – for the table's heading.
2. **Selected classes** – for selecting instrument classes used in orders to be shown in the table. If it is required to display only certain instruments, highlight a corresponding class in the list and use **Securities filter**.
3. **Firms filter, Depo accounts filter, Clients filter** – used to adjust different tables for different groups of clients or different market sectors.
4. **Color settings** – for setting the line background and font color relative to order execution status. For more detail see [3.5](#).
5. **Status filter** (Active, Filled, Canceled) – used to show in the table only orders having the indicated status.

6. **Only partially executed** – used to show in the table only partly filled orders in which **Quantity** is not equal to **Balance**.
7. **Operation filter** (Buy/Sell) – used to create a table containing transactions of one direction.
8. **Parameters set** – for selecting parameters to be shown in table's columns:

Parameter

Description

* Number	Order registration number in the exchange's trading system
----------	--

Parameter	Description
* Exchange code	Exchange ID
Trading date	Date of the current trading session
Date	Order registration date
* Sent (time)	Time of registering an order in the trading system
Sent (mcs)	Number of microseconds in time of order registration
* Period	Trading session period. Valid values: – Opening; – Trading; – Closing
Activation time	Order activation time
Killed (time)	Time of cancelling an order in the trading system
Killed (mcs)	Number of microseconds in time of order cancellation
Security (s.n.)	Short name of an instrument
* Security	Instrument name
Security code	Financial instrument code in the trading system
Class	Name of a class to which the instrument belongs
Class code	Class code in the trading system
* Side	Transaction direction (Buy / Sell)
* Depo account	Code of a trading account in which the order is entered
* Price	Order price per unit of an instrument
* Qty	Quantity of securities expressed in lots
* Balance	Volume of the unfilled part of an order expressed in lots
* Value	Volume of an order in terms of cash (excluding the exchange commission and accrued income)
Currency	Price currency, e.g. SUR for Russian Ruble
Yield	Yield as a percentage estimated for the order price
Accrued interest	Accumulated coupon interest calculated for a number of securities stated in the order in terms of cash
Trader	ID of a trader who entered the order
Dealer	ID of a firm on whose behalf the order has been entered

Parameter	Description
UID	User's ID at the QUIK server
Client code	Code of a client for which a limit of funds has been set
*** Comment	Supplementary reference information (entered by the trader). If the client code contains a forward slash (/), the value has the format <client_code>/<note to an algo order> ; if it does not, the format is <client_code>//<note to an algo order>
Linked order	Number of source order
Expiration	Time of filling an order
Type	Order type code of three characters: _ 1st character (order type): L – limited, M – market; _ 2nd character (price differentiation): S – at different prices, H – all trades at one price; _ 3rd character (execution condition): K – fill or kill, Q – insert in queue, W – kill remain
* Status	Order status (Active, Filled, Killed)
Trans ID	An individual number of transaction, TRANS_ID, used for transactions imported from a file
Settlement code	Code of payments under transaction. A parameter applicable to negotiation and REPO orders
Ransom price	Price of the REPO's second part per unit of an instrument. A parameter applicable to negotiation and REPO orders
Market-maker order	Denotes an order entered by a market maker
Reject reason	Trading system response to transaction

* – parameters set as default.

** – Comment field can take one of the following values depending on broker settings:

- _ algo order comment;
- _ algo order number;
- _ ID of user who entered algo order.

3.4 Available functions

The table data are available for copying into the Windows Clipboard and exporting through the DDE server and ODBC.

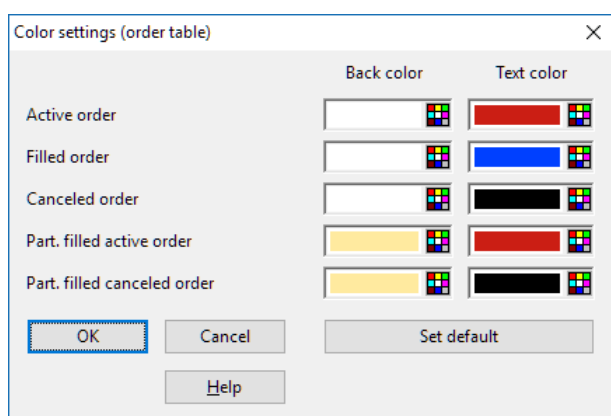
- Double left click of the mouse is to enter a new order with conditions similar to one on which the cursor is set;
- Double right click of the mouse is to withdraw an active order on which the cursor is set;
- 'F2' – to enter a new order;
- 'Ctrl+D' – to withdraw an order.

The above table contains only those orders that have been generated by the given algorithmic order. If a new order is entered manually using the linked order as a template, it will be displayed in the Table of Orders.

The complete list of control keys for all types of tables is shown in Annex to Section 2 of the QUIK User's Manual. Functions available for this table can be called from the shortcut menu of the table or the **Action** program menu (if the table is active). For the detailed description, see Section 3 of the QUIK User's manual, p. 3.6.4.

3.5 Color settings of linked orders

The relevant window can be called by clicking on button '...' right of the **Color settings** checkbox in the edit dialog box of the **Linked Orders** Table.



This function allows setting the background and font color for table lines relative to order execution status:

- **Active order** is for orders whose status is Active;
- **Filled order** is for orders whose status is Filled;
- **Killed order** is for orders whose status is Killed;
- **Part. filled active order** is for active orders that have been partly filled;
- **Part. filled killed order** is for cancelled orders that have been partly filled.

The button **Set default** returns the default settings shown in the picture.

4. Linked negotiated deal orders

4.1 Purpose of the table

The table **Linked negotiated deal orders. Algo order <Order number>** is intended for viewing orders entered for NDM classes and generated by a certain algorithmic order that have been registered in the trading system in the current trading session.

4.2 Table format

The table contains a chronologically arranged list of linked orders on negotiated deals. Each linked order is represented by a row in the table. Each parameter described in [4.3](#), pos. 8 is shown in an individual column.



Number	Opposite quote	Date	Sent(time)	Security (s.n.)	Security code	Class
1	1 192 690	03.10.2016	18:02:50	LUKOIL	LUKOIL [BK Ne LKOH	BK Negoti

4.3 Table settings

The table's edit dialog box contains the following settings:

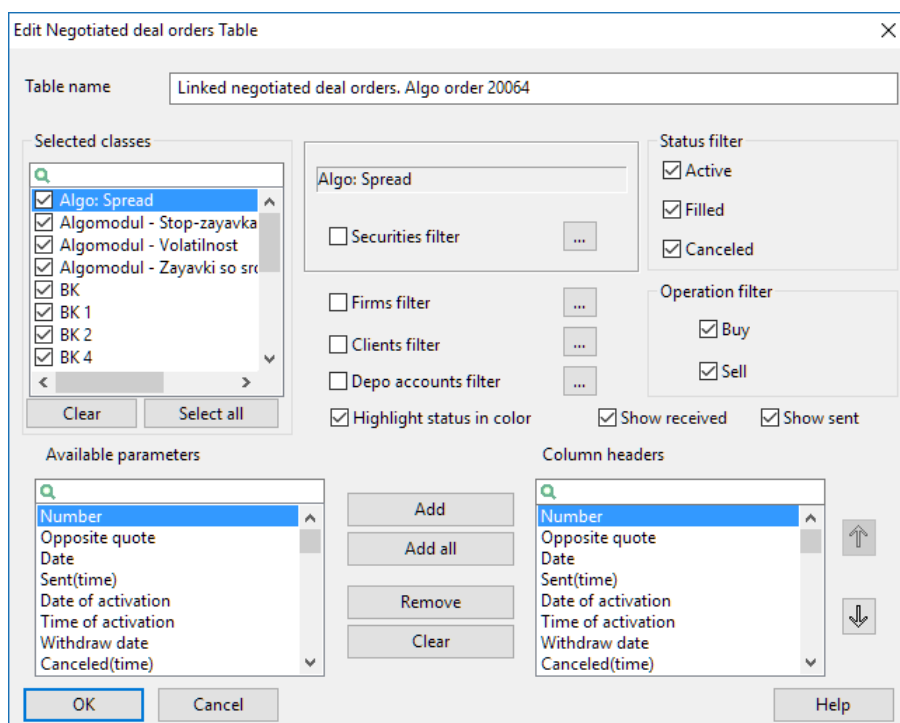


Table name: Linked negotiated deal orders. Algo order 20064

Selected classes:

- ☒ Algo: Spread
- ☒ Algomodul - Stop-zayavka
- ☒ Algomodul - Volatilnost
- ☒ Algomodul - Zayavki so sredstvami
- ☒ BK
- ☒ BK 1
- ☒ BK 2
- ☒ BK 4

Available parameters:

- Number
- Opposite quote
- Date
- Sent(time)
- Date of activation
- Time of activation
- Withdraw date
- Canceled(time)

Column headers:

- Number
- Opposite quote
- Date
- Sent(time)
- Date of activation
- Time of activation
- Withdraw date
- Canceled(time)

Filters:

- Algo: Spread
- ☐ Securities filter
- ☐ Firms filter
- ☐ Clients filter
- ☐ Depo accounts filter
- ☒ Highlight status in color
- Status filter:
 - ☒ Active
 - ☒ Filled
 - ☒ Canceled
- Operation filter:
 - ☒ Buy
 - ☒ Sell
- ☒ Show received
- ☒ Show sent

Buttons: Add, Add all, Remove, Clear, OK, Cancel, Help

1. **Table name** – for the table's heading.

2. **Selected classes** – for selecting instrument classes used in orders to be shown in the table. If it is required to display only certain instruments, highlight a corresponding class in the list and use **Securities filter**.
3. **Firms filter, Depo accounts filter, Clients filter** – used to adjust different tables for different groups of clients or different market sectors.
4. **Status filter** (Active, Filled, Canceled) – used to show in the table only orders having the indicated status.
5. **Operation filter** (Buy/Sell) – used to create a table containing transactions of one direction.
6. **Show received *** – display the orders with the Received value in the Direction status field.
7. **Show sent *** – display the orders with the Sent value in the Direction status field.

| (*) Use these settings to display sent and received orders in different tables.

8. **Highlight status in color** – if the check box is checked, the font in the table rows is highlighted depending on the Status field value:

- Active – red;
- Filled – blue;
- Cancelled – black.

If the check box is disabled, the font color is determined by operating system settings.

9. **Parameters set** – for selecting parameters to be shown in table's columns:

Parameter	Description
Number	Order registration number in the exchange's trading system
Opposite quote	Number of the counter quote. If it is absent, the value is 0
*** Date	Order registration date in the trading system
*** Sent (time)	Time of registering an order in the trading system. The format depends on the operating system settings
*** Canceled (time)	Time of cancelling an order in the trading system (for orders with Canceled status). The format depends on the operating system settings
*** Date of activation	Order activation date
*** Time of activation	Order activation time. The format depends on the operating system settings
*** Withdraw date	Date the order is withdrawn in format <YYYYMMDD>
Security (s.n.)	Short name of an instrument
Security	Instrument name

Parameter	Description
Security code	Instrument code in the trading system
Class	Name of a class to which the instrument belongs
Class code	Class code in the trading system
Side	Transaction direction (Buy / Sell)
Account	Depo account of a trading participant to whom the order is entered
Price	Order price per unit of an instrument
Qty	Quantity of securities in an order, expressed in lots
Balance	Volume of the unfilled part of an order expressed in lots
Trader	ID of a trader who entered the order
Dealer	ID of a trading participant who entered an order
Trader's org.	Name of a trading participant who entered an order
UID	User's ID at the QUIK server
Trader of partner	ID of the trader's firm to whom the order is entered. The field is not used and is empty
Partner	ID of the trading participant to whom the order is entered
Partner's org.	Name of the trading participant to whom the order is entered. The field is populated only for clients with active operations rights
Client code	Code of a client specified in an order
Comment	Supplementary reference field displaying the client code and a text comment of an order, separated by "/"
Reference	Text reference to explicitly identify two counter negotiated orders
Status	Order status. Valid values: – Active – unfilled order; – Filled – filled order; – Canceled – order is withdrawn by sender
Settlement code	Code of payments under transaction
Direction status	Order direction. Valid values: – Sent – personal order addressed to another trading participant; – Received – order of another trading participant addressed to you; – Sent and received – order addressed to yourself
Yield	Yield as a percentage estimated for the order price

Parameter	Description
Value	Volume of an order in terms of cash
Ransom price	Price of the REPO's second part per unit of an instrument, in terms of cash. For orders by SWAP instruments – basic rate of a currency instrument, specified by user when sending a transaction
Accrued profit	Coupon rate of interest on bonds, in monetary terms
Refund rate (%)	Refund rate in case of refusal to perform a trade, in % p.a.
* REPO rate (%)	REPO interest rate, in % p.a.
Transaction ID	An individual number of order, TRANS_ID, used for orders imported from a file
** REPO sum	REPO sum – sum of raised / borrowed funds by REPO trade, as at the current date
** REPO ransom value	REPO repurchase trade volume, in monetary terms
** Order input type	REPO order entry type. Valid values: – Price1+Rate; – Rate+Price2; – Price1+Price2; – REPO sum+Qty; – REPO sum+Discount; – Qty+Discount; – REPO sum; – Qty
** REPO period	REPO period in calendar days
** Initial discount(%)	Initial discount, in %
** Lower discount(%)	Lower value of the discount, in %
** Upper discount(%)	Upper value of the discount, in %
** Block securities	Indicates, that a financial instrument is frozen on the special account during the REPO transaction (Yes, No)
REPO sum original	Initial value of the REPO sum specified during the registration. Only for orders of modified REPO, for others (or if the value is not specified) – empty
Qty original	Initial number of securities (in standard lots) specified during the registration. Only for orders of modified REPO, for others (or if the value is not specified) – empty
Discount(%) original	Initial value of discount specified during the registration. Only for orders of modified REPO, for others (or if the value is not specified) – empty

Parameter	Description
Currency	Code of settlement currency, e.g. SUR for Russian Ruble, USD for USA dollar
On behalf of	The individual in whose name and from whose funds the trade is registered. Valid values: – On their own behalf, on their own expenses; – On their own behalf, at the expense of the client; – On their own behalf, at the expenses of trust management; – On behalf of the client, at the expense of the client
Large trade	Indicates that the trade is large (Yes/No)
Reg. number	State registration number of a security
*** Settle date	Settlement of the off exchange trade
Settle currency	Settlement currency code for a trade generated by this order
Account id	Account identifier in NCC (settlement code)
Linked order	Number of source order
Preferable security	Priority instrument in the collateral
UID canceled order	Code of the user cancelled an order on the QUIK server
System reference	Additional information on an order received from the trading system

* – this parameter is used only in orders for REPO trades.

** – parameter of orders for REPOs with state securities.

*** – if the setting **Show date and time of the trading data considering the local time zone** is enabled (menu **System/Settings/General settings...**, **Program** tab), then the value will be displayed with regard to time zone of the computer where the QUIK workstation is launched.

4.4 Available functions

The table data are available for copying into the Windows Clipboard and exporting through the DDE server and ODBC.

- Double left click of the mouse is to enter a new order with conditions similar to one on which the cursor is set;
- Double right click of the mouse is to withdraw an active order on which the cursor is set;
- F2 – to enter a new order;
- Ctrl+D – to withdraw an order.

The above table contains only those orders that have been generated by the given algorithmic order. If a new order is entered manually using the linked order as a template, it will be displayed in the Table of Negotiated deal orders.

The complete list of control keys for all types of tables is shown in Annex to Section 2 of the QUIK User's Manual. Functions available for this table can be called from the shortcut menu of the table or the **Action** program menu (if the table is active). For the detailed description, see Section 3 of the QUIK User's manual, p. 3.22.4.

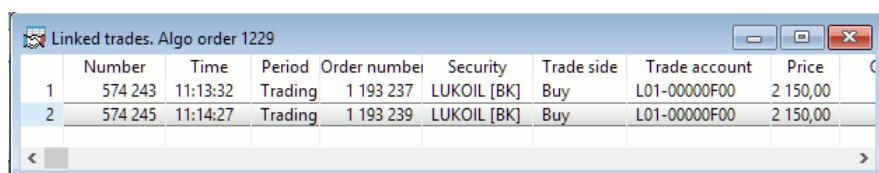
5. Linked trades

5.1 Purpose of the table

The table **Linked trades. Algo order <Order number>** is intended for displaying trades resulting from orders generated by a certain algorithmic order that have been registered with the trading system in the current trading session.

5.2 Table format

The table contains a chronologically arranged list of trades. Each trade is represented by a line in the table. Each parameter described in [5.3](#), pos. 6 is shown in an individual column.



The screenshot shows a window titled "Linked trades. Algo order 1229". Inside is a table with the following data:

	Number	Time	Period	Order number	Security	Trade side	Trade account	Price
1	574 243	11:13:32	Trading	1 193 237	LUKOIL [BK]	Buy	L01-00000F00	2 150,00
2	574 245	11:14:27	Trading	1 193 239	LUKOIL [BK]	Buy	L01-00000F00	2 150,00

5.3 Table settings

The table's edit dialog box contains the following settings:

1. **Table name** – for the table's heading.
2. **Selected classes** – for selecting instrument classes used in orders to be shown in the table. If it is required to display only certain instruments, highlight a corresponding class in the list and use **Securities filter**.
3. **Firms filter**, **Depo accounts filter**, **Clients filter** – used to adjust different tables for different groups of clients or different market sectors.
4. **Type filter** – used to filter transactions denoted as marginal. Transactions of a certain type are displayed in the table if the checkbox for that type is selected and are not if the checkbox is cleared.
5. **Operation filter** (Buy/Sell) – used to create a table containing transactions of one direction.
6. **Parameters set** – for selecting parameters to be shown in table's columns:

Parameter	Description
* Number	Order registration number in the exchange's trading system
* Exchange code	Exchange ID
Trading date	Date of the current trading session
Trade date	Date of registering a transaction

Parameter	Description
Settlement date	Date of payments under transaction
* Time	Time of registering a transaction in the trading system
Time (mcs)	Number of microseconds in time of order registration
* Period	Trading session period. Valid values: _ Opening; _ Trading; _ Closing
Order number	Number of an order being a basis for the transaction
Security (s.n.)	Short name of an instrument
* Security	Instrument name
Security code	Financial instrument code in the trading system
Class	Name of a class to which the instrument belongs
Class code	Class code in the trading system
Trade type	Margin trade indicator. If an outcome of the trade changes the current limit value for the client, the trade is designated as one of marginal type, otherwise, this field remains empty
* Trade side	Trade direction (Buy/Sell)
* Trade account	Code of a trading account in respect of which the trade has been made
* Price	Trade price per unit of an instrument
* Qty	Quantity of securities expressed in lots
* Value	Trade volume in terms of cash
Currency	Price currency, e.g. SUR for Russian Ruble
Settlement currency	Currency of trades settlement on MOEX
Settle code	Code for payments under negotiation deals
Yield	Yield as a percentage estimated for the transaction price
Accrued profit	Accumulated coupon interest calculated for a number of securities in the trade in terms of cash
Trader	ID of a trader who has made the trade
Dealer	ID of a firm on whose behalf the transaction has been made
Trader's org.	Trader' firm ID

Parameter	Description
Client code	Code of a client for which a limit of funds has been set
UID	User code on QUIK server
*, ** Broker reference	Supplementary reference information (entered by the trader). If the client code contains a forward slash (/), the value has the format <client_code>/<note to an algo order>; if it does not, the format is <client_code>//<note to an algo order>
Partner	ID of a trader with whom the transaction has been entered into (only for negotiation deals)
Partner's org.	ID of a firm with whom the transaction has been entered into (only for negotiation deals)
Ransom price	Buyback price of the REPO's second part in terms of cash
REPO rate	REPO interest rate, annual
TS Commission	Trading system commission charged on a transaction. For MOEX trades: TS Commission = Clearing center comission + Exchange commission + Trading center comission
Clearing center commission	Clearing fee. A parameter applicable to MOEX trades
Exchange commission	Stock exchange fee. A parameter applicable to MOEX trades
Trading center commission	Technical center fee. A parameter applicable to MOEX trades
Station ID	ID of a participant in MOEX trading
Profit (%) for ransom date	Interest income from a trade as of the buyback date in rubles
REPO sum	REPO amount is an amount of funds raised/provided for a REPO transaction as of the current date
REPO ransom value	REPO buyback value in rubles
REPO period	REPO term in calendar days
Start discount (%)	Initial discount as a percentage
Lower discount (%)	Lower discount rate limit as a percentage
Upper discount (%)	Upper discount rate limit as a percentage
Block securities	Flag of blocking a financial instrument on a special account for a period of REPO transaction (Yes/No)

Parameter	Description
Kind of trade	Trade type. Parameter of trades on the Moscow Exchange. Valid values: _ Regular trade; _ Targeted trade; _ Initial placement; _ Cash / securities transfer; _ Negotiated trade for REPO first leg; _ Settlement trade for swap operation; _ Settlement trade for OTC swap operation; _ Settlement trade of dual currency basket; _ Settlement OTC trade of dual currency basket; _ Settlement for REPO trade with CCP; _ First leg of REPO trade with CCP; _ Second leg of REPO trade with CCP; _ Negotiated trade for REPO with CCP; _ First leg of negotiated REPO trade with CCP; _ Second leg of negotiated REPO trade with CCP; _ Technical trade on returning REPO with CC assets

* – parameters set as default.

** – Broker reference field can take one of the following values depending on broker settings:

- _ algo order comment;
- _ algo order number;
- _ ID of user who entered algo order.

5.4 Available functions

The table data are available for copying into the Windows Clipboard and exporting through the DDE server and ODBC.

- Double left click of the mouse is to enter a new order with conditions similar to one on which the cursor is set;
- Double right click of the mouse is to make a chart;
- 'F2' – to enter a new order;
- 'F6' – to enter a new stop order.

The complete list of control keys for all types of tables is shown in Annex to Section 2 of the QUIK User's Manual. Functions available for this table can be called from the shortcut menu of the table or the **Action** program menu (if the table is active).). For the detailed description, see Section 3 of the QUIK User's manual, p. 3.8.4.

6. Algo orders limitation

The Module supports two types of algo order limitation:

- Limit for active algo orders – control of the summary current non-executed volume of all the active algo orders;
- Limit for a session – control of algo orders volume – executed and active at the moment – during the whole session.

Limitation is applied to GTD/GTC algo orders.

6.1 Limit set up

Limit creation dialog box can be opened using the following methods:

- by **Action / Set limit for cash** menu item selection if the Cash limits table is active;
- by clicking the **Set limit for cash** command from the shortcut menu in the Cash limits table.

The following parameters should be set up in the opened dialog box:

1. **Available firms** is the list of trading participant identifiers. Valid value: ALGO.
2. **Currency code** is the currency code for algo order settlements. SUR means Russian rubles, USD – US dollars. Select a code from the list.
3. **Group** is the identifier of trading session (trading mode), in which the limit is used. Valid values:
 - **ALOL** – limit for active algo orders;
 - **ALSL** – limit for a session.
4. **Client code** is the client code for which the limit is specified.
5. **Limit kind** is the type of the limit. Valid value: T0.
6. **Client position** is the value of client's own cash.
7. **Client limit** is not used.

Cash limit can be modified or deleted (for further details see [QUIK User's manual](#), Section 7, sub-section 7.12).

6.2 Cash table

After the limit is specified for a client it appears in the Cash table as a new record with the following parameters*:

Parameter	Description
Firm	Trading participant identifier: Algo
Currency	Currency code for algo order settlements
Group	Valid values: – ALSL; – ALOL
Client code	Client code for whom the limit is specified
Limit kind	Limit kind: T0
Incoming position	Valid values: – Specified limit for the session, for limits for a session; – Specified limit for active algo orders balance volume, for limits for active algo orders
Incoming limit	Valid values: – Summary volume balance (in cash) in all the active algo orders of the current client code and Current limit, for limits for a session; – Summary volume balance (in cash) in all the active algo orders of the current client code, for limits for active algo orders
Current position	The value is calculated according to the formula: Incoming position – Incoming limit
Current limit	Summary executed volume (in cash) in all the active algo orders of the current client code

* – other parameters of the table are not used for algo order limitation.

6.3 Work with limits

The value of **limit for active algo orders** balance decreases for an algo order volume on its registration, increases on its withdrawal or execution. At this algo order volume is calculated on the bases of algo order price.

The value of **limit for session** balance decreases for an algo order volume on its registration, increases on its withdrawal. On partial or full algo order execution limit balance increases for the value equal to the difference between trade volume calculated on the bases of algo order price and real trade volume.

For limit balance calculation on algo order registration the algo order volume must be calculated. Volume of algo orders with non-null price and quantity is calculated in a standard way.

If as a result of an algo order registration limit balance becomes lower than some alarm value specified by the administrator or limit balance becomes negative then a corresponding warning message is displayed on the workstation of the user who entered the order.

7. Entry and withdrawal of algorithmic orders

7.1 Methods of entering algorithmic orders

The algorithmic order input box may be activated by any of the following ways:

- Selecting the program menu item **Action / New algo order / [order type]** if the Algo orders table is active;
- Selecting the shortcut menu item **New algo order / [order type]** in the Algo orders Table;
- Selecting the shortcut menu item **New algo order / [order type]** in the Level II Quotes Table;
- Double clicking the left mouse button on a relevant line in the Algo orders Table;
- Pressing 'F2' on a selected line in the Algo orders Table.

7.2 Cancelling an active algorithmic order

An active algorithmic order may be cancelled by:

- Selecting the program menu item **Action / Kill algo order** if the Algo orders table is active;
- Selecting the shortcut menu item **Kill algo order** in the Algo orders Table;
- Double clicking the right mouse button on a relevant line in the Algo orders Table;
- Pressing 'Ctrl+D' in the Algo orders Table.

If an algorithmic order is cancelled, all active linked orders are withdrawn from the trading system.

7.3 Replacing an active algorithmic order

Operation of modifying an algorithmic order is available for of GTD/GTC-type algo orders. Modification can be executed by any of the following ways:

- Selecting the program menu item **Action / Edit algo order** if the Algo orders table is active;
- Selecting the shortcut menu item **Edit algo order** in the Algo orders Table;
- Pressing Ctrl+A in the Algo orders Table.

As a result, the dialog box for changing algo order conditions is shown on the screen:

- For GTD/GTC algo orders the following boxes are available:
 - Time in force;
 - Expiry date;
 - Expiry time;
 - Work interval;
 - Price;
 - Quantity (lot).

Dialog box titled "Edit algo order 20064" with the following fields and values:

- Class: BK Negotiation trades: A1-Shares 1 pai
- Firm: NC0038900000
- Instrument: Mecel
- Trading account: L01-00000F00
- Order type: Buy (unselected), Sell (selected)
- Time in force: Good Till Cancel
- Expiry date: 04.10.2016
- Expiry time: 10:42:01
- Work interval: ☐ (unchecked)
- Start: 0:00:00, End: 0:00:00
- Price: 100,000
- Quantity (lot 100): 1
- Settlement code: B01
- Market: ☐ (unchecked)
- Partner: (empty)
- Client code: Q3
- Comment: //123
- Alert on linked order execution: ☐ (unchecked)
- Buttons: Enter, Cancel

Click **Enter** to enable changes in an active order. Linked order with old parameters will be cancelled and resent only if price and/or quantity changed. In other situations, conditions of parent algo orders will be changed without resending a linked order.

- For Spread algo orders the following boxes are available for editing:
 - Time in force;
 - Expiry date;
 - Expiry time;
 - Spread type;
 - Spread;
 - Value;

- Iteration time, sec;
- Quantity/Value;
- Quantity in iteration;
- Value in iteration.

Click **Enter** to save changes in an active order. An active linked order is cancelled and then entered again with new conditions.

Replacement of other types of algorithmic order is possible solely by means of cancelling the active algo order and entering a new one.

7.4 Resuming execution of cancelled algo order

A cancelled algo order may be turned to active by any of the following ways:

- Selecting the program menu item **Action / Resume algo order** if the Algo orders table is active;
- Selecting the shortcut menu item **Resume algo order** in the Algo orders Table.

Execution of cancelled GTD/GTC, Spread and Stop algo orders may be resumed only if it is not expired. For Volatility algo orders that is possible only if an order is sent during current trading session.

8. Volatility-type order

8.1 Algorithm

This algorithm implies an opportunity to buy/sell options relative to a preset volatility value.

When an order is entered, the desired value of volatility and the maximum allowable deviation from volatility in percent or in volatility items are set. Then, the option price is calculated for the preset volatility value using the Black–Scholes model, and the order is entered into the trading system at the calculated price. Upon each change in parameters taken into account in the calculation of price with respect to volatility, a maximum and minimum allowable option price is calculated regarding the allowable deviation. If the price of the entered order falls beyond that range, the unfilled quantity is reentered with a new price.

As the intermediate clearing begins, all active volatility orders are automatically suspended. If a new linked order is entered at this time, the clearing status for the instrument is checked (parameter **clstate**). If the status is **Int. clearing active**, the algo order execution is suspended. When attempting to remove an active volatility order, the system generates a diagnostics message: 'Impossible to remove the algo order N... Int. clearing active'.

As the intermediate clearing is accomplished, the algo order continues to operate.

Formulas

Theoretical price of an option (Black–Scholes model):

$$V_{call} = SN(d_1) - Ke^{-rT} N(d_2), V_{put} = N(-d_2)Ke^{-rT} - SN(-d_1)$$

$$d_1 = \frac{\left[\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T \right]}{\sigma\sqrt{T}}, d_2 = d_1 - \sigma\sqrt{T},$$

where

- V_{call} is the theoretical price for a 'call' option;
- V_{put} is the theoretical price for a 'put' option;
- S is the current price of the underlying asset (for FORTS options the **Calculated price** parameter is used);
- T is the time before expiry;
- K is the option's strike;
- r is the risk-free interest rate;
- N is the standard distribution function;
- e is the Napierian base (2.71828);

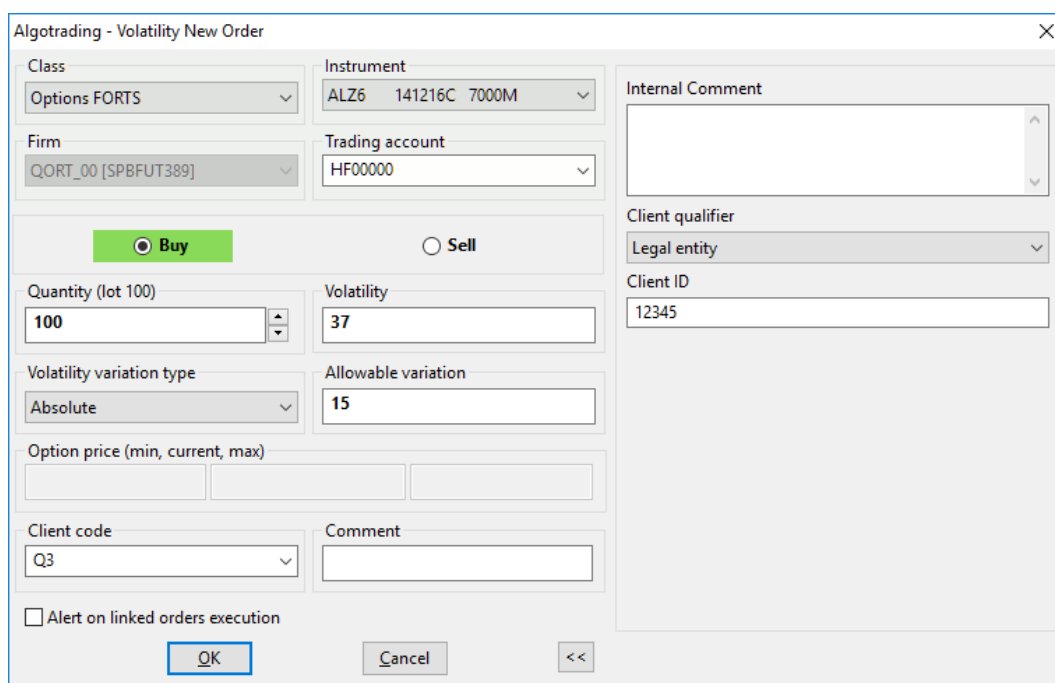
- σ is the volatility.

8.2 Order entry

The window for entering a Volatility order may be activated by any of the following ways:

- Selecting the program menu item **Action / New algo order / Algotrading – Volatility** if the Algo orders table is active;
- Selecting the shortcut menu item **New Algo order / Algotrading – Volatility** in the Algo orders Table;
- Selecting the shortcut menu item **New algo order / Algotrading – Volatility** in the Level II Quotes Table.

Click the  /  buttons to expand / collapse the additional panel of Volatility New Order entry dialog box.



Parameters stated in the order are:

1. **Class** – to select a class to which the traded instrument belongs.
2. **Instrument** – to select an instrument from the list of contracts of that class.
3. **Firm** stands for ID of a firm participating in trade on whose behalf the order is registered.
4. **Trading account** – a trading account for transaction.
5. **Buy/Sell** – to determine the transaction direction.
6. **Quantity (lot ...)** – a number of contracts expressed in lots. In parentheses, the number of securities in one lot is shown.
7. **Volatility** – to select the allowable volatility value (σ).
8. **Volatility variation type** – type of volatility variation estimation:

- **% from volatility** – volatility deviation is set in percent;
- **Absolute** – volatility deviation is set in absolute items.

9. Allowable variation – a volatility deviation allowed.

10. Option price (min, current, max) – reference boxes showing option price values corresponding to the volatility and allowable variation values that have been set. The bottom option price (left box) corresponds to the volatility value with an allowable variation downward, the current price (middle box) to the preset volatility value, and the top price (right box) the volatility value with an allowable variation upward. Values set in these boxes are recalculated once the volatility and allowable variation values that have been set or the last traded price for the option's basic asset change.

Showing option prices in these boxes requires that:

- The box Calculate price by volatility in algo order entry dialog box be checked in the program settings (menu item **System/Settings/General settings...**, section **Trading / Orders / Algo orders**);
- Receiving the **Last trade price** parameter is set for all futures being options' basic assets. If the method **According to settings for open user tables** is selected in the program settings (menu item **System/Settings/General settings...**, section **Program / Data receiving**), the receipt of parameter will be set automatically. If another method is selected, one should open the menu item **System/Data request/Available securities...**, select the class **Futures FORTS** and ensure that **Last trade price** is present in the filter of received parameters. The receipt of this parameter should not be disabled; otherwise, the option prices calculation would be incorrect.

11. Client code – the client's ID in QUIK.

12. Comment – a text comment to an order.

13. Alert on linked orders execution – the enabled checkbox means that alert will be sent on linked orders execution.

14. Internal Comment – text comment for internal use. The comment is not sent to the trading system. The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

15. Client qualifier – client qualifier according to the MiFID II directive. Valid values:

- No qualifier;
- Natural Person;
- Legal Entity.

The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

16. Client ID – client identifier according to the MiFID II directive. The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

9. GTD/GTC orders

9.1 Algorithm

The algorithm of a GTD/GTC order involves the simulation of the exchange order with the expiration dates Good Till Date, Good Till Cancel or Good Till Time.

When registering a GTD/GTC order a linked order with the following parameters is immediately or during the Work interval (if specified) placed on the exchange:

- expiration date – Session;
- type – Limited or Market.

Other parameters of a linked order correspond to the parameters of an algo order placed by a user.

When cancelling a GTD/GTC order a linked order is cancelled. When cancelling a linked order a GTD/GTC order is cancelled.

When the Module starts up in the beginning of the day the unexpired GTD/GTC order places a linked order on the exchange.

Good Till Time orders cancel a linked order after a specified in the algo order entry dialog box time and change its status to **Cancelled**.

9.2 Order entry

The window for entering a GTD/GTC order may be activated by any of the following ways:

- Selecting the program menu item **Action / New algo orders / Algotrading – GTD/GTC orders** if the Algo orders table is active;
- Selecting the shortcut menu item **New algo order / Algotrading – GTD/GTC orders** in the Algo order Table;
- Selecting the shortcut menu item **New algo order / Algotrading – GTD/GTC orders** in the Level II Quotes Table.

Click the  /  buttons to expand / collapse the additional panel of GTD/GTC New Order entry dialog box.

Algotrading: GTD/GTC orders New Order

Class MOEX SM: T+ A1-Shares, pai i RDR	Firm IKSI [NC0038900000]	Internal Comment
Instrument LUKOIL	Trading account L01-00000F00	
☒ Buy ☐ Sell		Client qualifier Natural person
Time in force Good Till Cancel	Expiry date 17.11.2017	Client ID 156
	Expiry time 17:49:51	
☐ Work interval Start: 7:04:02 End: 10:02:03		
Price 1000,000	Quantity (lot 100) 10	Settlement code
☐ Market	max: 10000	
Partner NC0038900000 [NC0038900000]		
Client code Q3	Comment	
☐ Alert on linked order execution		
Enter Cancel <<		

Parameters stated in the order are:

1. **Class** – to select a class to which the traded instrument belongs.
2. **Firm** stands for ID of a firm participating in trading on whose behalf a linked order has been registered.
3. **Instrument** – to select an instrument from the list of securities of that class.
4. **Trading account** – a trading account available for transaction.
5. **Buy / Sell** – direction: to buy or to sell.
6. **Time in force** – time period during which an order has to be executed. Valid values:
 - _ Good Till Time;
 - _ Good Till Date;
 - _ Good Till Cancel.
7. **Expiry date *** – date (included) till which an algo order must be filled. The parameter is available if the values Good Till Time or Good Till Date are specified in the **Time in force** parameter.
8. **Expiry time *** – time of the end of order execution. The parameter is available if the Good Till Time value is specified in the **Time in force** parameter.
9. **Work interval** – time period during which the exchange order must be active:
 - _ **Start **** – time of the exchange order sending;
 - _ **End **** – time of the exchange order cancel.

If the option is selected, the order is placed on exchange every day in time specified in the **Start** parameter and cancelled in time specified in the **End** parameter within its validity period.

Whether the Work interval parameter is required to be filled or not is set up by administrator in the server part of the Module.

10.Price – instrument unit price.

11.Market – attribute of a linked market order.

12.Quantity (lot ...) – total quantity of securities in this algo order, in lots. Quantity of securities in one lot is specified in brackets.

13.max: – maximum number of lots in an order calculated on the basis of the client's funds including all commission fees. Click the button to enter the maximum value to the **Quantity (lot...)** box.

14.Settlement code – settlement code of the operation. The box is available if the **Class** box contains the NDM class.

15.Partner – trader's ID who entered an order. The box is available if the **Class** box contains the NDM class.

16.Client code – the client's ID in QUIK.

17.Comment – a text comment to an order.

18.Alert on linked orders execution – send a notification when a linked order is executed.

19.Internal Comment – text comment for internal use. The comment is not sent to the trading system. The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

20.Client qualifier – client qualifier according to the MiFID II directive. Valid values:

- No qualifier;
- Natural Person;
- Legal Entity.

The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

21.Client ID – client identifier according to the MiFID II directive. The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

(*) If the Show date and time of the trading data considering the local time zone parameter is enabled (see QUIK User's manual, Section 2, sub-section 2.10.1) then the values of Expiry date and Expiry time parameters are displayed according to the time zone of the computer where the QUIK terminal is running.

()**

1. **Default values of the beginning time and the end time when exchange orders cancelled for the certain exchange class are set by administrator in settings of the Module server part.**
2. **If values of the start and the end of activity coincide, the order is rejected.**

10. Algorithmic stop order

10.1 Algorithm

The algorithm is aimed to imitate a stop order intended to close a client position on spot and derivatives markets.

Available validity periods of an algorithmic Stop order:

- Session;
- Good till time;
- Good till date;
- Good till cancel.

If the exchange price of an instrument specified in Stop order is different from a user-defined corridor around client's volume-weighted average price, all active orders sent for this instrument are cancelled (if the option **Cancel client orders** is active) and a limited or market linked order is sent to close a client's current position.

When sensing a Stop order is possible to set the time interval during which Stop order controls price of the last transaction and client's position within each trading session.

Size of a client position and volume-weighted average price is calculated by server, the Module only uses this information.

To get information on a position for derivatives market instruments use the data from Securities limits table. Size of a client position is equal to the value of Current balance field, volume-weighted average price is equal to the value of WA.position price field.

To get information on positions for instrument of spot-market use the data from Client account positions (futures) table. Size of a client position is equal to the value of Cur. pure pos. field, volume-weighted average price is equal to the value of Effect. pos. price field.

Only one Stop order can be sent on a client position.

Execution

The following additional terms are introduced:

- **LastAllTradePrice** – price of the last whole trade for an instrument;
- **PositionAvgPrice** – volume-weighted average price of a client position for an instrument. For spot market instruments it is the value of WA.position price field in Securities limits table. For instruments of derivatives market it is the value of Effect. pos. price field in Client account positions (futures) table;
- **TakeProfit** – offset of the exchange price from volume-weighted average price of a position to be closed by Take-profit;
- **SafeSpread** – value of the protective spread. Protective spread sets an additional (advanced) offset of an order price from 'broken through' corridor boundary and intended to make a sent order deliberately executable;
- **HighBound** – high boundary of the available range;
- **LowBound** – low boundary of the available range;
- **CheckPosition** – position in lots which is controlled by a Stop order;
- **ClientPosition** – a client position. For instrument of spot market it is equal to the value of Current balance in Securities limits table. For instrument of derivatives market it is equal to the value Cur. pure pos. field in Client account positions (futures) table;
- **LotSize** – lot size of an instrument;
- **ActiveBalance** – unfilled quantity in a linked order. Sign defines the direction of a linked order:
 - _ '-' – buy;
 - _ '+' – sell.

Process of a Stop order execution after successful registration is described below.

Step 1

Step 1 is to control a client position.

If the value of **CheckPosition** is different from 0 then subscription to all trades is executed. Otherwise Stop order expects a client position to be closed.

When occurring one of the following events the algorithm passes to the step 2:

- Whole trade is received;
- Information on a client position is refreshed.

Step 2

Step 2 is to control temporary limits and activation conditions to send a linked order.

If Take-profit and Stop-limit are not set then an appropriate condition (Take-profit / Stop-limit) is not accounted by Stop order.

Activation conditions:

LastAllTradePrice >= HighBound

or

LastAllTradePrice <= LowBound,

where:

_ If ClientPosition > 0 a linked order is sent to sell:

_ Take-profit is set in percent:

HighBound = PositionAvgPrice * (1 + TakeProfit / 100)

_ Take-profit is set in percent:

HighBound = PositionAvgPrice + TakeProfit

_ Stop-limit is set in percent:

LowBound = PositionAvgPrice * (1 - StopLimit / 100)

_ Stop-limit is set in absolute values:

LowBound = PositionAvgPrice - StopLimit

_ If ClientPosition < 0 a linked order is sent to buy:

_ Stop-limit is set in percent:

HighBound = PositionAvgPrice * (1 + StopLimit / 100)

_ Stop-limit is set in absolute values:

HighBound = PositionAvgPrice + StopLimit

_ Take-profit is set in percent:

LowBound = PositionAvgPrice * (1 - TakeProfit / 100)

_ Take-profit is set in absolute values:

LowBound = PositionAvgPrice - TakeProfit

When occurring one of the events the algorithm passes to the step 3.

Step 3

Step 3 is to send a linked order.

If Stop order has an active linked order then this order is killed.

If the option **Cancel client orders** is selected in Stop order dialog box then all active client orders sent for the specified instrument are killed.

A linked order is sent with a price of **Price** field value calculated depending on the direction and conditions of functioning according to the following formulas:

1. To sell:

- _ When functioning Take-profit:

- _ Protective spread is set in percent:

$$\text{Price} = \text{HighBound} * (1 - \text{SafeSpread} / 100)$$

- _ Protective spread is set in in absolute value:

$$\text{Price} = \text{HighBound} - \text{SafeSpread}$$

- _ When functioning Stop-limit:

- _ Protective spread is set in percent:

$$\text{Price} = \text{LowBound} * (1 - \text{SafeSpread} / 100)$$

- _ Protective spread is set in in absolute value:

$$\text{Price} = \text{LowBound} - \text{SafeSpread}$$

2. To buy:

- _ When functioning Take-profit:

- _ Protective spread is set in percent:

$$\text{Price} = \text{LowBound} * (1 + \text{SafeSpread} / 100)$$

- _ Protective spread is set in in absolute value:

$$\text{Price} = \text{LowBound} + \text{SafeSpread}$$

- _ When functioning Stop-limit:

- _ Protective spread is set in percent:

$$\text{Price} = \text{HighBound} * (1 + \text{SafeSpread} / 100)$$

- _ Protective spread is set in in absolute value:

$$\text{Price} = \text{HighBound} + \text{SafeSpread}$$

If the option **Market** is selected a linked order is sent at market price (**Price = 0**).

When sending a linked order **Qty** field value is expressed in lots. If positions are handled in pieces then a linked order is sent for quantity:

$$\text{Qty} = \text{abs}(\text{ClientPosition}) / \text{LotSize}$$

Otherwise, if a position is set in lots:

$$\text{Qty} = \text{abs}(\text{ClientPosition})$$

After sending a linked order to prevent double response of Stop order a client position balance **CheckPosition** is calculated and will be controlled by Stop order:

- in pieces:

$$\text{CheckPosition} = \text{ClientPosition} / \text{LotSize} - \text{ActiveBalance}$$

- in lots:

$$\text{CheckPosition} = \text{ClientPosition} - \text{ActiveBalance}$$

Then Stop order is unsubscribed from a trade and passes to the Step 1.

If a linked order was sent unsuccessfully (rejected transaction) the following events are processed:

- Session for the instrument is closed – Stop order is waiting for opening a session for this instrument;
- Connection failed – Stop order is waiting for reconnection to the Module;
- Other – Stop order is being killed.

Stop order with validity period **Good till time** when coming the time specified in an algo order kills the linked order and changes Stop order status to **Killed**.

When killing a Stop order a linked order is also killed.

Stop order is active till it is not killed by user otherwise it is not killed automatically according to validity period or when unsuccessfully sending a linked order.

10.2 Order entry

The window for entering a Stop order may be activated by any of the following ways:

- Selecting the program menu item **Action / New algo order / Stop order** if the Algo orders table is active;
- Selecting the shortcut menu item **New Algo order / Stop order** in the Algo orders Table;
- Selecting the shortcut menu item **New algo order / Stop order** in the Level II Quotes Table.

Click the  /  buttons to expand / collapse the additional panel of Stop order New Order entry dialog box.

Parameters stated in the order are:

1. **Class** – to select a class to which the traded instrument belongs.
2. **Firm** stands for ID of a firm participating in trade on whose behalf the order is registered.
3. **Instrument** – to select an instrument from the list of contracts of that class.
4. **Trading account** – a trading account for transaction.
5. **Time in force** – validity period of a Stop order. Valid values:
 - _ Before cancellation;
 - _ During the session;
 - _ Up to date;
 - _ Up to time.
6. **Expiry date *** – date Stop order execution. The box is available if **Time in force** value is Up to date or Up to time.
7. **Expiry time *** – time of a Stop order execution finishing. The box is available if **Time in force** value is Up to time.
8. **Watch interval** – time period during which a Stop order conditions are checked.
 - _ **Start *,***** – time when the Stop order begins controlling a client position;
 - _ **End *,***** – time when the Stop order finishes controlling a client position.
9. **Take-profit** – offset of exchange price from volume-weighted average price for closing on take-profit:

- Box for entering Take-profit values. The box is not available is the value 'Not set' is selected in **Take-profit type** box;
- Box **Take-profit type** **:
 - Money – in price currency;
 - Not set – value of Take-profit is not accounted in calculation of a Stop order price;
 - Percent – in percent.

10.Stop-limit – offset of exchange price from volume-weighted average price for closing on stop-limit:

- Box for entering Stop-limit values. The box is not available is the value 'Not set' is selected in **Stop-limit type** box;
- Box **Stop-limit type** **:
 - Money – in price currency;
 - Not set – value of Stop-limit is not accounted in calculation of a Stop order price;
 - Percent – in percent.

11.Cancel client orders – kill active client orders when closing a position.

12.Spread – value of the protective spread. Protective spread sets an additional (advanced) offset of an order price from 'broken through' boundary of the corridor and intended to make a sent order deliberately executable:

- Box of entry of the protective spread value;
- Box **Spread type**:
 - Money – in price currency;
 - Percent – in percent.

13.By market price – send a linked order at market price when closing a client position.

14.Client code – identifier of a client in QUIK system.

15.Comment – text comment to an order.

16.Alert on linked orders execution – send a notification when executed a linked order.

17.Internal Comment – text comment for internal use. The comment is not sent to the trading system. The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

18.Client qualifier – client qualifier according to the MiFID II directive. Valid values:

- No qualifier;
- Natural Person;
- Legal Entity.

The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

19. Client ID – client identifier according to the MiFID II directive. The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

(*) When setting Show date and time of the trading data considering the local time zone (see QUIK user's manual, sub-section 2.10.1) is active the values of parameters Expiry date, Expiry time, Start and End are specified with consideration of time zone of the computer where QUIK terminal is run.

() Take-profit type and Stop-limit type boxes cannot have the value 'Not set' at the same time.**

(*)**

- 1. Default values of the beginning time and the end time when the Stop order begins controlling a client position are set by administrator in settings of the Module server part.**
- 2. If values of the start and the end of activity coincide, the stop order is rejected.**

11. Algorithmic spread order

11.1 Algorithm

This algorithm implies that one instrument is bought and another is sold while the spread is kept (not less than minimum set by the user) within the buy and sell prices of instruments.

At the algorithm input, the following parameters are set:

- **Security_{buy}** – an instrument to be bought by the algorithm;
- **Security_{sell}** – an instrument to be sold by the algorithm;
- **Value** – total volume of an algo order;
- **Currency** – currency in which the volume is set;
- **SpreadType** – the spread type (absolute or relative);
- **Spread** – spread value. If the spread type is Absolute, the value is shown in terms of cash; if the spread type is Relative, the value is shown in the ratio;
- **IterQtyValue** – iteration size type (quantity or volume);
- **IterValue** – the volume the linked algo orders are registered for if the iteration size type is Volume;

- **IterQty** – lot quantity to be fulfilled during the one iteration if the iteration size type is Quantity;
- **IterTime** – iteration maximum time.

Additional terms:

- **ValueBalance_{buy}** – volume available for purchase;
- **ValueBalance_{sell}** – volume available for sale.

Additionally, the validity period on an algo order and the time period during which it remains active are set.

Each of the following conditions must be met in order for an algo order to work correctly:

- Non-zero demand for an instrument $Security_{sell}$;
- Non-zero supply for an instrument $Security_{buy}$;
- Open trading session for the instruments $Security_{sell}$ and $Security_{buy}$.

Spread algo orders operate iteratively. The volume which is not more than IterValue (IterQty) is filled at each iteration. Each iteration includes the following steps.

Step 1

Order registration.

Calculation of the orders' prices:

- For buy orders:

– Spread type is absolute:

$$P_{buy} = bestbid_{Security_{sell}} - Spread$$

– Spread type is relative:

$$P_{buy} = bestbid_{Security_{sell}} * Spread$$

- For sell orders:

– Spread type is absolute:

$$P_{sell} = bestoffer_{Security_{buy}} + Spread$$

– Spread type is relative:

$$P_{sell} = \frac{bestoffer_{Security_{buy}}}{Spread}$$

Calculation of the orders' quantity:

- For buy orders:

- _ Iteration size type is volume:

$$Qty_{buy} = \min \left(\text{rounddown} \left(\frac{IterValue}{P_{buy} * Lot} \right), \frac{ValueBalance_{buy}}{P_{buy} * Lot} \right)$$

- _ Iteration size type is quantity:

$$Qty_{buy} = \min \left(IterQty, \frac{ValueBalance_{buy}}{P_{buy} * Lot} \right)$$

- For sell orders:

- _ Iteration size type is volume:

$$Qty_{sell} = \min \left(\text{rounddown} \left(\frac{IterValue}{P_{sell} * Lot} \right), \frac{ValueBalance_{sell}}{P_{sell} * Lot} \right)$$

- _ Iteration size type is quantity:

$$Qty_{sell} = \min \left(IterQty, \frac{ValueBalance_{sell}}{P_{sell} * Lot} \right)$$

If the volume is set by the user in a currency which is different from the instrument currency, the volume will be calculated in the instrument currency adjusted for the cross rate received from the QUIK server in the CROSSRATE class.

When a zero quantity is calculated for any of the orders, the algo order will be cancelled (if $IterValue < P$) or gone to the step 4.

Otherwise, the algo order registers two linked limit orders with the calculated parameters and goes to the step 2.

Step 2

Waiting for order execution.

The order is pending execution of any linked order for **IterTime** seconds, and then goes to the step 3.

If any of the orders is fully executed, the order is no more pending and goes to the step 3.

Step 3

At this step, a decision to proceed with attempts to fill the unfilled balance of volume/quantity in the current iteration (time interval) or transfer the balance to the next iteration is made.

The Level II Quotes table is analyzed, and if there are no changes the orders pending execution (they are not cancelled).

If both orders are fully executed, a new iteration will begin (step 1).

If one of the limit orders (order №1) is fully executed and the opposite (order №2) is active, then the order №2 will be cancelled and a market order with the same parameters and price equal to 0 will be entered to execute the unfilled balance of the order №2. A new iteration begins (step 1).

If both orders are active but one of these orders (№1) is partly executed and its balance is less than the opposite order's balance (№2), both orders will be cancelled. And then, a market order to execute the volume calculated as the difference between the executions of orders №1 and №2 is entered. If the quantity calculated depending on the executed volume is equal to 0, a market order will not be entered. A new iteration begins (step 1).

Step 4

At this step, the final stage of the algo order execution takes place.

If the quantities calculated for both directions at the step 1 are equal to 0, the algo order receives the Filled status. And then, the algo order execution process finishes.

Otherwise, the algo order puts a limit order of the direction in which the quantity is calculated as not equal to 0 at the price calculated using the formulas:

- For buy orders:

- Spread type is absolute:

$$P_{buy} = P_{Avg_{sell}} - Spread$$

- Spread type is relative:

$$P_{buy} = P_{Avg_{sell}} * Spread$$

- For sell orders:

- Spread type is absolute:

$$P_{sell} = P_{Avg_{buy}} + Spread$$

- Spread type is relative:

$$P_{sell} = \frac{P_{Avg_{buy}}}{Spread}$$

Where

- $P_{Avg_{sell}}$ is a weighted average price of the linked sell orders' execution;
- $P_{Avg_{buy}}$ is a weighted average price of the linked buy orders' execution.

An algo order remains active until a linked limit order is filled or cancelled.

If a linked order is executed, an algo order receives the Filled status. And then, the algo order execution process finishes.

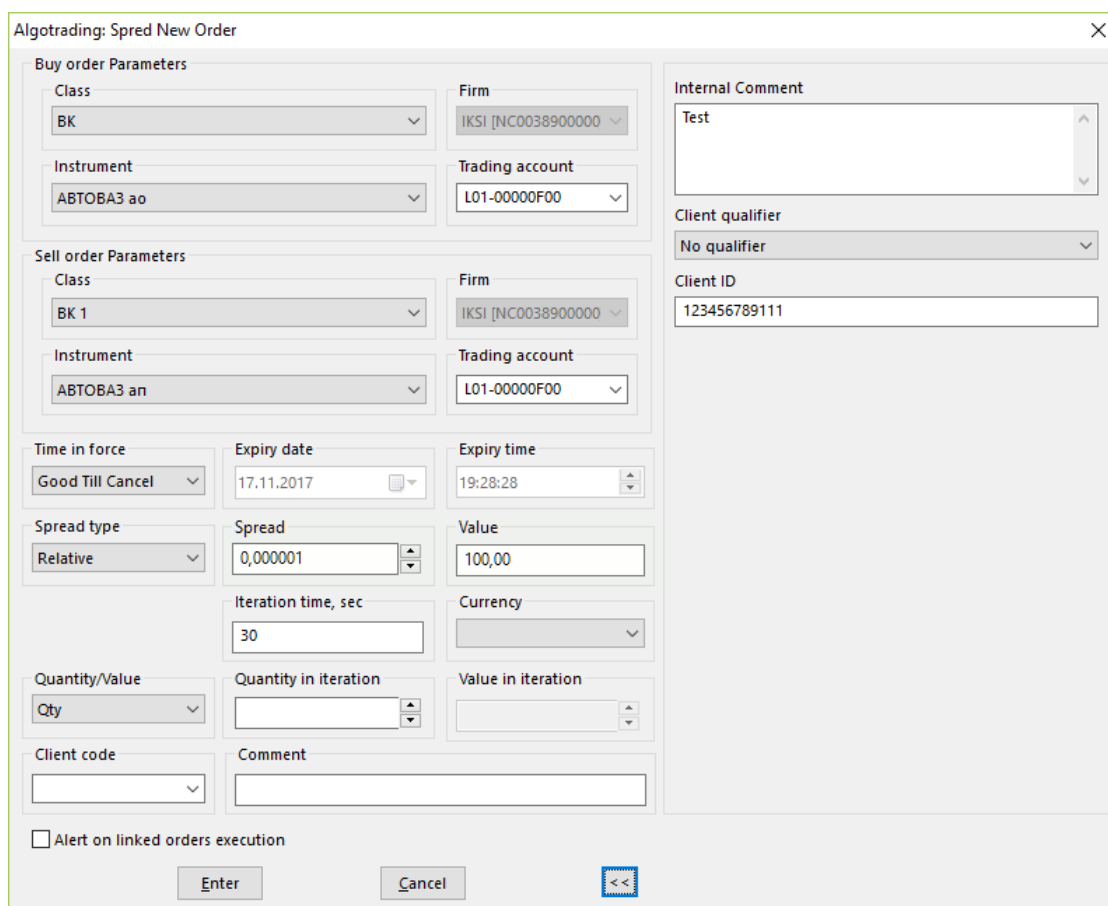
■ **Spread algo order remains 'live' during the main trading mode.**

11.2 Order entry

The window for entering a Spread order may be activated by any of the following methods:

- Selecting the program menu item **Action / New algo order / Spread** if the Algo orders table is active;
- Selecting the shortcut menu item **New Algo order / Spread** in the Algo orders Table;
- Selecting the shortcut menu item **New algo order / Spread** in the Level II Quotes Table.

Click the  /  buttons to expand / collapse the additional panel of Spred New Order entry dialog box.



Parameters stated in the order are:

1. Buy order parameters:

- **Class** allows selecting a class to which the traded instrument belongs;

- **Instrument** allows selecting an instrument from the list of contracts of that class;
- **Firm** stands for ID of a firm participating in trade on whose behalf the order is registered;
- **Trading account** – a trading account for transaction.

2. Sell order parameters:

- **Class** allows selecting a class to which the traded instrument belongs;
- **Instrument** allows selecting an instrument from the list of contracts of that class;
- **Firm** stands for ID of a firm participating in trade on whose behalf the order is registered;
- **Trading account** – a trading account for transaction.

3. Time in force – time period during which an order has to be executed. Valid values:

- Good Till Cancel;
- Good Till Date;
- Good Till Time;
- Session.

4. Expiry date * – date (included) till which an algo order must be filled. The box is available if **Time in force** value is Good Till Date or Good Till Time.

5. Expiry time * – time of an algo order execution finishing. The box is available if **Time in force** value is Good Till Time.

6. Spread type:

- Absolute – in terms of cash;
- Relative – in the ratio.

7. Spread – safe spread value.

8. Value – volume of an algo order.

9. Iteration time, sec allows to set the duration in seconds of the one iteration of an algo order for automatic calculation of the iterations number.

10. Currency – currency in which the volume is set.

11. Quantity/Value – the method of showing the amount of linked orders in each iteration.

12. Quantity in iteration – quantity (in units) to be fulfilled during the one iteration. The box is available if the value of the Qty/Value parameter is Qty.

13. Value in iteration – volume the linked algo orders are registered for during the one iteration. The box is available if the value of the Qty/Value parameter is Value.

14. Client code – identifier of a client in QUIK system.

15. Comment – text comment to an order.

16. Alert on linked orders execution – send a notification when executed a linked order.

17. Internal Comment – text comment for internal use. The comment is not sent to the trading system. The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

18. Client qualifier – client qualifier according to the MiFID II directive. Valid values:

- No qualifier;
- Natural Person;
- Legal Entity.

The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

19. Client ID – client identifier according to the MiFID II directive. The visibility of the box is determined by the administrator with the help of the settings of the server part of the Module.

(*) Default values of the beginning and end time of the algorithm operation are set by administrator in settings of the Module server part.

12. Requesting a report

12.1 Report on an algo order

The report is intended for formal presentation of results achieved by a selected algorithmic order. Reports may be requested by any of the following ways:

- Selecting the menu item: **Action / Get report** if the Algo orders table is active;
- Selecting the shortcut menu item **Get report...** in the Algo orders Table.

Report on SPREAD algo order # 20095

Report on SPREAD algo order # 20095

Registration Time

30.09.2016 12:57:20

Execution alerts

No

Time in Force

Good till cancel

Client Code

Q3

Spread Type

Absolute

Spread Value

100.000000 SUR

Value

150000.00 SUR

Qty in Iteration

3

Iteration Time

30

Status

Active

Buy

Sell

Class

BQUOTE

BQUOTE

Instrument

LKOH

SBER

Firm ID

NC0038900000

NC0038900000

Trade Account

L01-00000F00

L01-00000F00

Balance (Value)

150000.00 SUR

150000.00 SUR

Execution VWAP

0.00 SUR

0.00 SUR

Transactions

Time

ID

Order Number

Status

Type

Side

Price

Quantity

Class

Instrument

Trade Account

Message

The report's header contains the name and number of an algo order being the subject of report. In the header, the following information is presented the parameters specified by a user are displayed in the upper part of algo order parameters table and the dynamic parameters of an algo order are displayed in the lower part (status, execution parameters):

Parameter	Description
Side	Transaction direction: Buy / Sell. The parameter is applicable to Volatility, GTD/GTC and Spread algo orders (for each instrument in a pair)
Registration time	Time of registering an algo order in QUIK
Instrument	Instrument code, class code. For Spread algo orders – instrument code (for each instrument in a pair)
Class	Class code. Spread algo order parameter (for each instrument in a pair)
Trade account	Trading account number. This parameter is applicable to Stop algo orders

Parameter	Description
Firm ID	Firm identifier. Spread algo order parameter (for each instrument in a pair)
Execution alerts	Indicates that alert will be sent on linked orders execution
Time in Force	A time period for an order to be filled. Applicable to GTD/GTC, Stop and Spread algo orders
Client code	Client code in the QUIK server
Qty	Total number of securities in a given algo order expressed in lots. For Stop algo orders: quantity in active linked order
Balance	Unfilled part of an algo order expressed in lots. This parameter is applicable to Volatility and Stop orders
Status	Current order execution status: Active, Filled, Cancelling or Cancelled
Kill time	Time of cancelling an algo order
Executed quantity	An executed part of algo order in lots
Executed value	An executed part of algo order in terms of cash. Applicable to TWAP/VWAP orders, Stop orders
Execution VWAP	Volume weighted average fill price in cash. This parameter is applicable to Stop algo orders
Execution VWAP (price)	Volume weighted average fill price in pips. This parameter is applicable to VWAP orders for derivatives market instruments
Type	Order type: Limit or Market. This parameter is applicable Stop orders and GTD/GTC orders
Price	Price per unit of the instrument shown in an algorithmic order
Volatility	Volatility value. This parameter is applicable to Volatility orders
Volatility deviation	Allowable deviation percentage. This parameter is applicable to Volatility orders
Type of Volatility Deviation	— Absolute – volatility deviation is set in absolute items; — % from volatility – volatility deviation is set in percent. This parameter is applicable to Volatility orders
Cancel Client Orders	Attribute of cancelling client orders when closing a position (Yes / No). This parameter is applicable to Stop algo orders
Use time limit	Attribute of using the activity interval. This parameter is applicable to Stop algo orders
Start Watch time	Time of the activity start. This parameter is applicable to Stop algo orders

Parameter	Description
Stop Watch time	Time of the activity end. This parameter is applicable to Stop algo orders
Safe Spread Type	– Money – Safe Spread percent parameter is not filled; – Percent – Safe Spread parameter is not filled. This parameter is applicable to Stop algo orders
Spread	Value of the protective spread expressed in terms of cash. The parameter is not filled if the protective spread type is 'Percent' or 'Not set'. This parameter is applicable to Stop algo orders
Spread percent	Value of the protective spread expressed as a percentage. The parameter is not filled if the protective spread type is 'Money' or 'Not set'. This parameter is applicable to Stop algo orders
Spread Type	– Absolute – the value is shown in terms of cash; – Relative – the value is shown as the ratio. This parameter is applicable to Spread algo orders
Spread value	Spread value. The value is shown in terms of cash or as the ratio depending on the value of the Spread Type parameter. This parameter is applicable to Spread algo orders
Value	Volume of an algo order in terms of cash (excluding the exchange commission and accrued income)
Qty in Iteration	Lot quantity to be fulfilled during the one iteration. The parameter is not filled if the value of the Qty/Value parameter is Value. This parameter is applicable to Spread algo orders
Value in Iteration	Volume the linked algo orders are registered for during the one iteration. The parameter is not filled if the value of the Qty/Value parameter is Qty. This parameter is applicable to Spread algo orders
Safe Spread percent	Value of the protective spread in percent. The parameter is not filled if the protective spread type is 'Money' or 'Not set'. This parameter is applicable to Stop algo orders
Take-profit Type	– Money – Take-profit percent parameter is not filled; – Not set – Take-profit and Take-profit percent parameters are not filled; – Percent – Take-profit parameter is not filled. This parameter is applicable to Stop algo orders
Take-profit	Value of take-profit in price currency. The parameter is not filled if the Stop-limit type is 'Percent' or 'Not set'. This parameter is applicable to Stop algo orders
Stop-limit Type	– Money – Stop-limit percent parameter is not filled; – Not set – Stop-limit and Stop-limit percent parameters are not filled; – Percent – Stop-limit parameter is not filled.

Parameter	Description
	This parameter is applicable to Stop algo orders
Stop-limit percent	Value of Stop-limit in percent. The parameter is not filled if the protective spread type is 'Money' or 'Not set'. This parameter is applicable to Stop algo orders
Counterparty firm ID of Linked Orders	Code of the firm entered a linked order
Settlement Code	Settlement code

Reporting tables contain the following information:

- **Transactions** – a list of transactions to enter/cancel linked orders generated by an algorithmic order with indication to their execution status;
- **Linked orders** – a list of linked orders registered in the trading system;
- **Linked trades** – a list of filled linked trades.

Once the report is received, it can be viewed in a pop-up window on the screen. The **Message Window** will notify of the time at which the report has been received. Each received report is opened in a new window.

The report can be saved as a HTML file or printed out. These functions are available from the shortcut menu in the report viewing window.

12.2 Report on an algo order for a previous period

The report is intended for formal presentation of results achieved by a selected algorithmic order for a previous period. Reports may be requested by any of the following ways:

- Selecting the menu item: **Action / Get report for previous period ...** if the Algo orders table is active;
- Selecting the shortcut menu item **Get report for previous period...** in the Algo orders Table.

In the **Report of algo order** window opened the algorithm and number of an algo order and a client code are specified.

If the value specified in the **Algorithm** or **Client code** boxes does not match the order number specified in the **Number of algo order** box, an appropriate diagnostics message is displayed.

The report's header and tables contain the information on the algo order. For the description, see [12.1](#).